

Pilot Study

Data Naming Convention & Folder Structure Proposal

Multi-Device Physiological Monitoring Pipeline

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1. Overview

This document proposes a standardized data naming convention and folder structure for the multi-device physiological monitoring pilot study. The goal is to ensure consistency, reproducibility, and scalability as the project moves from pilot testing to full data collection.

The design follows two core principles:

1.

Separation of concerns: REDCap handles participant metadata (demographics, consent, EMA); the pipeline handles device-level physiological data. Both systems are linked through a shared ID scheme.
2.

De-identification by design: No personally identifiable information (PII) exists in the pipeline data. Only coded participant IDs and session IDs are used.

2. Identifier System

A consistent ID system is the bridge between REDCap and the device data pipeline. All IDs are purely numeric codes with no embedded PII.

2.1 Participant ID

Format: **sub-XXX** (e.g., sub-001, sub-002, ..., sub-999)

This ID is assigned sequentially when a participant is enrolled. The same ID must be used in both REDCap and the pipeline file system. The mapping between sub-XXX and the participant's real identity is stored only in REDCap's access-controlled environment.

2.2 Session ID

Format: **ses-XXX** (e.g., ses-001, ses-002)

Each continuous monitoring period (e.g., a 24-hour wear cycle) is one session. Sessions are numbered sequentially per participant.

2.3 ID Correspondence

System	Participant ID	Session ID	Contains PII?
REDCap	sub-XXX	ses-XXX	Yes (access-controlled)
Pipeline	sub-XXX	ses-XXX	No
Analysis	sub-XXX	ses-XXX	No

3. Device Registry

Each device is assigned a short, lowercase identifier used consistently in folder names and file names.

Device ID	Device Name	Primary Signals	Notes
polar	Polar H10	ECG, HRV	Chest strap, Bluetooth
empatica	Empatica EmbracePlus	EDA, HR, Temp, Accel	Wrist-worn, cloud export
muse	Muse 2	EEG	Headband, Bluetooth
respiban	respiBAN	Respiration, ECG, IMU	Chest sensor, OpenSignals
masimo	Masimo MightySat	SpO2, Pulse	Finger clip, manual export
geneactiv	GENEActiv	Actigraphy, Light	Wrist-worn (not yet available)
ontrak	OnTrak ABPM	Blood Pressure	Cuff-based (not yet available)
ema	EMA (Phone)	Subjective reports	Via REDCap, exported as CSV

4. Folder Structure

The root project directory is organized into functional subdirectories. Each level has a clear purpose.

4.1 Top-Level Structure

`project_root/`

```
raw_data/           # Original device exports, never modified
processed_data/     # Parsed & time-normalized data
aligned_data/       # Multi-device aligned outputs
quality_reports/    # Data quality check results
configs/            # Pipeline configuration files
scripts/            # All code
docs/               # SOPs, protocols, this document
logs/               # Pipeline execution logs
```

4.2 Raw Data Directory

Raw data is organized by participant, then session, then device. Original files are placed as-is without renaming.

```
raw_data/

  sub-001/

    ses-001/

      polar/          <- Polar H10 exports

      empatica/       <- Empatica exports

      muse/           <- Muse 2 exports

      respiban/       <- respiBAN exports

      masimo/         <- Masimo exports

      ema/            <- EMA CSV from REDCap

      session_log.json <- Session metadata

    ses-002/

      ...

  sub-002/

    ...
```

4.3 Session Log File

Each session directory contains a **session_log.json** that records metadata about the recording session:

```
{

  "participant_id": "sub-001",

  "session_id": "ses-001",

  "start_time_utc": "2026-03-20T14:00:00Z",

  "end_time_utc": "2026-03-21T14:00:00Z",

  "timezone": "America/New_York",

  "devices_used": ["polar", "empatica", "muse", "respiban", "masimo"],
```

```

    "notes": "",

    "operator": "ZG"

}

```

5. File Naming Convention

5.1 Raw Data Files

Raw files exported from devices are NOT renamed. They are placed into the correct device folder as-is to preserve provenance.

5.2 Processed Data Files

After parsing and time normalization, files follow a standardized naming pattern:

```
{participant_id}_{session_id}_{device_id}_{signal_type}.parquet
```

File Name	Description
sub-001_ses-001_polar_ecg.parquet	Polar H10 ECG data
sub-001_ses-001_empatica_eda.parquet	Empatica EDA data
sub-001_ses-001_empatica_hr.parquet	Empatica heart rate
sub-001_ses-001_muse_eeg.parquet	Muse 2 EEG data
sub-001_ses-001_respiBAN_resp.parquet	respiBAN respiration
sub-001_ses-001_masimo_spo2.parquet	Masimo SpO2 data
sub-001_ses-001_ema_responses.parquet	EMA survey responses

5.3 Unified Data Format

All processed Parquet files share the same column structure:

Column	Type	Description
timestamp_utc	datetime64[ns, UTC]	UTC timestamp, the universal time axis
timestamp_local	datetime64[ns]	Local time (timezone from session_log)
value	float64	Signal value (unit depends on signal type)

quality_flag	int8	0 = valid, 1 = suspect, 2 = invalid
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6. Data Quality Rules

Each parser module applies basic quality checks during processing. Records that fail are flagged but not deleted, preserving all original data.

Flag	Condition	Example	Action
0 (valid)	Passes all checks	Normal reading within range	Use in analysis
1 (suspect)	Physiologically unlikely	HR > 220 bpm, SpO2 < 70%	Review before use
2 (invalid)	Technical failure	NaN, disconnect, timestamp gap	Exclude from analysis

7. Pipeline Architecture

The pipeline is modular with four sequential stages. Each reads from the previous stage's output, ensuring reproducibility.

Stage	Name	Input	Output
1	Parse	raw_data/{sub}/{ses}/{device}/	processed_data/{sub}/{ses}/
2	Quality Check	processed_data/{sub}/{ses}/	quality_reports/{sub}/{ses}/
3	Align	processed_data/{sub}/{ses}/	aligned_data/{sub}/{ses}/
4	Export	aligned_data/{sub}/{ses}/	Final analysis-ready dataset

Adding a new device in the future only requires writing a new parser module (Stage 1). Stages 2-4 operate on the unified format and require no modification.

8. REDCap Integration

The pipeline does not interact with REDCap directly. The connection is maintained through the shared ID scheme:

3. When a participant is enrolled in REDCap, they receive a sub-XXX ID.
4. The same sub-XXX is used to create their folder in raw_data/.
5. EMA data collected via REDCap is exported as CSV and placed in the ema/ subfolder.

6. At analysis time, pipeline outputs merge with REDCap exports using participant_id + session_id as join keys.

This design ensures that PII (stored only in REDCap) is never mixed with device data in the pipeline.

9. Open Questions for Discussion

7. Participant ID format: Is sub-XXX (3-digit) sufficient, or should we use sub-XXXX (4-digit)?
8. Session definition: For 24-hour continuous wear, does one calendar day = one session, or is a session defined by device on/off cycles?
9. EMA export frequency: Will EMA data be exported from REDCap per-session, per-participant, or in bulk?
10. Storage location: Should the project_root be on a shared drive, local machine, or BU research computing?
11. Multi-channel handling: For devices with multiple signals (e.g., Empatica: EDA + HR + Temp + Accel), separate files per signal or one combined file?